MARINE AI INTELLIGENCE: COMPLETE PROJECT REPORT

Automated Marine Debris & Ghost Net Acoustic Signature Engine

Smart India Hackathon (SIH 2026) · Ministry of Earth Sciences (MoES)

Document Ref: MoES/SIH-2026/MAI-REP-v1.0	**Status:** Full Pipeline Validated & Live
Date: September 6, 2026	**Target Architecture:** Simulator + Spring Boot + React Dashboard

1. Executive Summary

Marine debris, specifically Abandoned, Lost, or otherwise Discarded Fishing Gear (ALDFG / 'Ghost Nets'), represents an urgent ecological and maritime navigation hazard. Side-scan sonar (SSS) survey operations have historically struggled with three primary limitations: (1) Black-box AI models that fail to provide acoustic rationale to sonar hydrographers; (2) High false alarm rates caused by natural seabed structures; and (3) Severe spatial resolution constraints where standard surveys cannot physically resolve net mesh.

Marine AI Intelligence provides an end-to-end, scientifically transparent acoustic detection suite. It combines a physics-forward acoustic simulation, a 5-channel geometric regularity engine, a high-throughput Spring Boot 3 REST API, and a modern interactive web dashboard featuring Explainable AI (XAI) and survey geometry advisories.

2. System Architecture & Components Built

Component	Tech Stack	Deliverable
Acoustic Simulator	Python 3.10+, NumPy, OpenCV, SciPy	Slant-range waterfall renderer with grazing angles, shadows, Rayleigh speckle, and 2x2 anti-aliased supersampling.
5 Signatures Engine	Python reference & Java production	Periodicity (FFT mesh pitch), Linearity (SVD), Circularity, Angularity, and Irregularity (rejection channel).
Production REST API	Spring Boot 3, Java 21, OpenCV	Endpoints (/api/analyze, /detections, /statistics) with real per-stage profiling (130ms end-to-end latency).
Synthetic Dataset	Physics Forward Model	3 benchmark survey swaths (sim_00000 to sim_00002) with YOLO labels, masks, and ground-truth metadata.
Interactive Dashboard	React 18, Vite, TypeScript, Tailwind v4	Live UI on http://localhost:5173 with pan/zoom canvas, metric distance ruler, XAI radar, and Nyquist calculator.

3. Core Scientific Innovation: The 5 Geometric Signatures

Core Scientific Thesis: *Manufactured objects are geometrically regular; nature is not.* The seabed contains no straight lines, no right angles, no radial circles, and no centimeter-scale repeating grids. The pipeline evaluates candidates across five mathematical channels:

- **Periodicity (Ghost Nets / Mesh):** 2D FFT radially averaged and whitened against 1/f red-noise baseline. Recovers mesh pitch (2.0-18.0cm) and infers commercial gear (Purse Seine, Gillnet, Trawl). Strong periodicity (>=0.45) decisively overrides boundary roughness.
- **Linearity (Cables, Pipes, Towlines):** Evaluates bounding rectangle elongation combined with SVD principal axis straightness, filtering out natural undulating sand ridges.
- **Circularity (Drums, Tyres, Buoys):** Evaluates isoperimetric quotient $4\pi A/P^2$ combined with centroid radial consistency.
- **Angularity (Containers, Crates):** Uses Douglas-Peucker polygon approximation to count interior angles near 90° (±15°) and edge straightness.
- **Irregularity (Natural Rejection Channel):** Negative control channel measuring boundary roughness and spectral flatness. Natural rocks score >0.70 and are rejected with zero false alarms.

4. Experimental Benchmark Validation & Performance

- **Accuracy:** 10/11 correct classifications on procedural benchmark.
- **False Alarm Rate on Natural Seabed:** 0.00% (zero false positives on rocks, coral heads, and sediment dunes).
- **Pipeline Latency:** 130.7 ms average compute time per 2048x1024 side-scan sonar image.

5. Scientific Limitations & Operational Guidelines

1. **Nyquist Resolution Limit:** Net mesh recovery requires across-track ground resolution dG <= P_mesh / 2.5. High-frequency surveys (H=3.5m, R=20m) achieve ~2cm/px and successfully resolve mesh. Standard surveys (H=15m, R=100m, ~10cm/px) cannot resolve mesh, and the system transparently reports this limitation.
2. **Linearity Curvature:** Taut ropes and rigid pipes score high linearity (>0.8), while slack or coiled cables exhibit higher RMS deviation.
3. **Physics Forward-Model Disclosure:** Synthetic forward-model data is openly disclosed due to the worldwide absence of public labelled ghost net sonar corpuses.